KAMAL ACHARYA

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ACADEMIC PROFILE

Postdoctoral Research Associate at Baylor University with a Ph.D. in Information Systems from the University of Maryland, Baltimore County (UMBC). Research focuses on neurosymbolic AI, trustworthy and interpretable machine learning, and learning-based optimization for intelligent decision-making, with applications in advanced air mobility and transportation systems. Published in IEEE Transactions, IJCAI, and other peer-reviewed venues, with research experience in NASA and USDOT-funded projects and over a decade of university-level teaching experience.

RESEARCH INTERESTS

Neurosymbolic AI for safety-critical decision systems; theory-guided and data-driven demand modeling; Advanced Air Mobility and intelligent transportation systems; reinforcement learning and optimization; trustworthy and interpretable AI.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Postdoctoral Research Associate Aug 2026 – Present
Baylor University, Waco, TX, USA

- Conduct research under the NASA University Leadership Initiative (ULI) project on mobility-energy coordinated infrastructure planning for Advanced Air Mobility (AAM) systems.
- Develop and evaluate optimization approaches for energy-aware eVTOL fleet operations, including aircraft assignment, passenger scheduling, and operational decision-making.
- Investigate centralized and distributed optimization methods, including mixed-integer optimization and the Alternating Direction Method of Multipliers (ADMM), for scalable fleet coordination.
- Design computational experiments and simulation-based evaluations to assess operational performance, energy constraints, and coordination strategies under varying demand scenarios.
- Integrate data-driven travel demand forecasting with fleet optimization and simulation to support AAM infrastructure planning and operational decision-making.

Graduate Research Assistant Aug 2023 – May 2026
University of Maryland, Baltimore County, Baltimore, MD, USA

- Conducted doctoral research under the NASA University Leadership Initiative (ULI) on mobility–energy coordinated infrastructure planning for Advanced Air Mobility (AAM).
- Developed scalable machine learning and neurosymbolic AI methods for transportation demand forecasting, trip generation, and regional air mobility planning using large-scale spatiotemporal datasets.
- Designed a rolling-horizon trip request generation framework for time-dependent AAM demand simulation, supporting infrastructure planning and operational decision-making.
- Developed interpretable predictive models integrating machine learning and symbolic reasoning for transportation demand estimation.
- Published multiple first-author papers in IEEE Transactions on Intelligent Transportation Systems, IJCAI, IEEE CAI, IEEE ITSC, IEEE Big Data, IWCMC, and AIAA based on this research.

Teaching Assistant
Operationalizing AI/Machine Learning for Cybersecurity Training May 2024 – Aug 2024
University of Maryland, Baltimore County (MD, USA)

- Supported a multidisciplinary graduate training program on Artificial Intelligence, Machine Learning, and Cybersecurity supported by NSF grants (OAC-2309760 and OAC-2229976).

- Assisted instructors in delivering hands-on instruction on AI-driven cybersecurity and cyberinfrastructure tools.
- Mentored interdisciplinary student teams conducting research projects integrating AI and cybersecurity applications.
- Provided guidance on project implementation, technical reports, and final research presentations.

Research Assistant Aug 2022 – Aug 2023
Embry-Riddle Aeronautical University, Daytona Beach, FL, USA

- Conducted research on machine learning and learning-based optimization for air mobility demand forecasting and emergency transportation planning.
- Contributed to the USDOT-funded Center for Advanced Transportation Mobility (CATM) project, with research activities continuing after transitioning to the University of Maryland, Baltimore County (UMBC).
- Developed deep learning models for air mobility prediction and investigated learning-to-optimize approaches for operational decision-making.

Information Technology Coordinator 2014 – 2021
College of Applied Business and Technology (Kathmandu, Nepal)

- Helped establish a research and development unit in the IT sector in the college.
- Taught multiple undergraduate computing and programming courses within the institution.
- Conducted guest lectures and special seminars; supervised and coordinated faculty duty rosters.
- Counseled students; hired IT-related faculty in consultation with leadership.
- Developed and administered non-credit IT courses; organized annual IT symposium/workshops and competitions.
- Encouraged student publications (magazines, newspapers, journals).

Web Developer 2013 – 2014
Uniweb Technologies (Kathmandu, Nepal)

- Developed and maintained websites/web applications using HTML, CSS, JavaScript, and PHP.
- Designed responsive interfaces for cross-device/browser compatibility.
- Managed and optimized databases using complex SQL queries; improved performance via refactoring and caching.
- Integrated third-party APIs (payment gateways, social media); performed unit/integration/UAT testing.

Visiting Lecturer 2012 – 2021
Various Colleges Affiliated to Tribhuvan and Purbanchal Universities (Kathmandu, Nepal)

- Taught: C, C++, Computer Organization, Computer Architecture, Digital Logic, Advanced Internetworking.
- Supervised major and minor projects; involved in BIM, BCA, and BIT programs.

Lecturer 2012 – 2013
Janakpur Engineering College (Kathmandu, Nepal)

- Delivered assigned courses; encouraged assignments and student counseling.
- Guided field work; engaged students in research/projects; prepared detailed course plans.
- Supervised minor and major projects.

EDUCATION

Ph.D., Information Systems May 2026
University of Maryland, Baltimore County (UMBC)
Dissertation: *Enhancing Demand Modeling for Advanced Air Mobility Using Data-Driven Learning and Neurosymbolic AI*
Advisor and Committee Chair: Dr. Houbing Song
Co-Advisor: Dr. Alvaro Velasquez
Committee Members: Dr. Nirmalya Roy, Dr. Lei Zhang, Dr. Liang Sun, Dr. Kai Sun

M.Sc., Information System Engineering 2019
Purbanchal University
Thesis: An Analysis of Classification Algorithms for Nepali News
Selected Coursework: Software Engineering, Object-Oriented Systems, Information Security, Advanced Database Management, Operations Research, Distributed and Cloud Computing, Data Mining and Warehousing, Data Communication and Computer Network.

Bachelor of Electronics and Communication Engineering 2011
Tribhuvan University
Final Year Project: Distance and Speed Measurement Using Ultrasound
Selected Coursework: Computer Programming I & II, Computer Architecture, Microprocessors, Logic Circuits, Digital Electronics, Computer Graphics, Numerical Methods, Probability and Statistics, Control Systems, Signal Analysis, Digital Signal Processing, Communication Systems I & II.

PUBLICATIONS

Journal Articles

- K. Vasiloff, **K. Acharya**, M. A. Arif, Z. Wang, H. H. Song, F. Wang, and L. Sun, "Demand Forecast and Energy-Aware Portal Siting for Regional Air Mobility," *Journal of Air Transport Management*, vol. 139, Art. no. 103143, 2026. [DOI]
- **K. Acharya**, W. Raza, K. Vasiloff, Z. Wang, L. Sun, and H. H. Song, "Demand Modeling for Advanced Air Mobility: Challenges, Opportunities, and Future Directions," *IEEE Transactions on Intelligent Transportation Systems*, vol. 27, no. 7, pp. 7454–7474, 2026. [DOI]
- **K. Acharya** and H. H. Song, "A Comprehensive Review of Neuro-symbolic AI for Robustness, Uncertainty Quantification, and Intervenability," *Arabian Journal for Science and Engineering*, vol. 51, no. 1, pp. 35–67, 2026. [DOI] **(Invited Paper)**
- A. R. L. de Macêdo, S. K. Jagatheesaperumal, K. A. P. da Costa, **K. Acharya**, H. H. Song, M. Guizani, and V. H. C. de Albuquerque, "Quantum AI-Enhanced IoT–Fog Communication: A Survey From Cybersecurity and Data Privacy Perspective," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 3637–3665, 2026. [DOI]
- **K. Acharya**, A. Velasquez, and H. H. Song, "A Survey on Symbolic Knowledge Distillation of Large Language Models," *IEEE Transactions on Artificial Intelligence*, vol. 5, no. 12, pp. 5928–5948, Dec. 2024. [DOI]
- **K. Acharya**, W. Raza, C. Dourado, A. Velasquez, and H. H. Song, "Neurosymbolic Reinforcement Learning and Planning: A Survey," *IEEE Transactions on Artificial Intelligence*, vol. 5, no. 5, pp. 1939–1953, May 2024. [DOI] **(IEEE CIS Publication Spotlight, Nov. 2024)**
- W. Raza, X. Ma, H. H. Song, A. Ali, H. Zubairi, and **K. Acharya**, "Long Short-Term Memory Neural Network Assisted Peak to Average Power Ratio Reduction for Underwater Acoustic Orthogonal Frequency Division Multiplexing Communication," *KSII Transactions on Internet and Information Systems*, vol. 17, no. 1, 2023. [DOI]

Conference Proceedings

- **K. Acharya**, H. H. Song, K. Vasiloff, Z. Wang, and L. Sun, "Urban Air Mobility Flight Demand Modeling for Airports in New York City," in *AIAA SCITECH 2026 Forum*, Paper AIAA 2026-1475, 2026. [DOI]
- **K. Acharya**, I. Sharifi, M. Lad, L. Sun, and H. H. Song, "Integrating Neurosymbolic AI in Advanced Air Mobility: A Comprehensive Survey," in *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 10362–10370, 2025. [DOI]
- **K. Acharya**, M. Lad, L. Sun, and H. H. Song, "Neurosymbolic Approach for Travel Demand Prediction: Integrating Decision Tree Rules into Neural Networks," in *2025 International Wireless Communications and Mobile Computing Conference (IWCMC)*, Abu Dhabi, UAE, pp. 600–605, 2025. [DOI]
- **K. Acharya**, M. Lad, L. Sun, and H. H. Song, "A Data-Driven Approach to Enhancing Gravity Models for Trip Demand Prediction," in *2025 IEEE Conference on Artificial Intelligence (CAI)*, Santa Clara, CA, USA, pp. 815–820, 2025. [DOI]

- **K. Acharya**, M. Lad, H. H. Song, and L. Sun, "Regional Air Mobility Flight Demand Modeling in Tennessee State," in *AIAA SCITECH 2025 Forum*, Paper AIAA 2025-2783, 2025. [DOI]
- S. B. Hakim, M. Adil, **K. Acharya**, and H. H. Song, "Decoding Android Malware with a Fraction of Features: An Attention-Enhanced MLP-SVM Approach," in *Network and System Security (NSS 2024)*, Lecture Notes in Computer Science, vol. 15564, Springer, 2025. [DOI]
- M. Lad, **K. Acharya**, L. Sun, and H. H. Song, "Enhancing Forecasting for Advanced Air Mobility," in *2024 IEEE International Conference on Big Data (BigData)*, Washington, DC, USA, pp. 2908–2913, 2024. [DOI]
- **K. Acharya**, M. Lad, L. Sun, and H. H. Song, "Demand Modeling for Advanced Air Mobility," in *2024 IEEE International Conference on Big Data (BigData)*, Washington, DC, USA, pp. 2855–2863, 2024. [DOI]
- **K. Acharya**, A. Velasquez, Y. Liu, D. Liu, L. Sun, and H. H. Song, "Improving Air Mobility for Pre-Disaster Planning with Neural Network Accelerated Genetic Algorithm," in *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, AB, Canada, pp. 1290–1295, 2024. [DOI]
- L. Zhu, Q. Lan, A. Velasquez, H. H. Song, **K. Acharya**, Q. Tian, and S. Niu, "SKGHOI: Spatial-Semantic Knowledge Graph for Human-Object Interaction Detection," in *2023 IEEE International Conference on Data Mining Workshops (ICDMW)*, Shanghai, China, pp. 1186–1193, 2023. [DOI]

Book Chapters

- S. B. Hakim, M. Adil, **K. Acharya**, and H. H. Song, "AI for Android Malware Detection and Classification," in *AI for Cybersecurity*, H. H. Song et al., Eds., Wiley, 2025. [DOI]

FUNDED RESEARCH PROJECTS

- **NASA University Leadership Initiative (ULI)**: Mobility-Energy-Coordinated Platform for Infrastructure Planning (MECPIP) to Support Advanced Air Mobility Operations. *Postdoctoral Research Associate, Baylor University (2026–Present).* Contributing to energy-aware fleet operations, distributed optimization, and AAM infrastructure planning. [Project Overview]
- **U.S. Department of Transportation (USDOT) – Center for Advanced Transportation Mobility (CATM)**: Machine Learning for Improving Air Mobility under Emergency Situations (Grant No. 69A3551747125). *Research Contributor.* Conducted research on machine learning-based air mobility prediction and optimization. [Final Report]
- **National Science Foundation (NSF) – CyberTraining Program**: Collaborative Research: Operationalizing AI/Machine Learning for Cybersecurity Training (Grants OAC-2309760 and OAC-2229976). *Graduate Teaching Assistant, UMBC (2024).* Supported interdisciplinary AI/ML and cybersecurity training and student research activities. [Training Syllabus]

HONORS & AWARDS

- Conference Travel Grant ($1000), Department of Information Systems, UMBC, 2026
- Professional Development Grant ($750), Graduate Student Association, UMBC, 2026
- Outstanding Reviewer (2024–2025), IEEE Transactions on Neural Networks and Learning Systems, IEEE Computational Intelligence Society, Dec 2025
- Doctoral Research Poster Award, UMBC, May 2025
- IEEE CIS Publication Spotlight Recognition, IEEE Computational Intelligence Society, Nov 2024

TEACHING EXPERIENCE AND MENTORSHIP

University Courses Taught

- **Programming and Software Development:** C Programming, C++, PHP, Python.
- **Artificial Intelligence and Data Science:** Artificial Intelligence, Data Mining.
- **Computer Systems and Architecture:** Computer Organization, Computer Architecture, Microprocessors (8085/8086), Real-Time Systems.
- **Computer Graphics and Visualization:** Computer Graphics.

- **Networking and Communications:** Data Communication, Computer Networks, Advanced Internetworking, Analog and Digital Communication.
- **Digital Systems:** Digital Logic Design.

Student Supervision and Mentorship

- Supervised undergraduate major and minor projects in computing and engineering, providing guidance on problem formulation, implementation, and technical documentation.
- Mentored interdisciplinary student teams during the NSF-funded AI/ML CyberTraining program, supporting project implementation, technical reports, and research presentations.

Curriculum Development and Pedagogy

- Developed instructional materials and assessments for undergraduate computing and engineering courses.
- Incorporated project-based learning, practical instruction, and applied examples into undergraduate teaching.

INVITED TALKS AND GUEST LECTURES

- "Neurosymbolic AI for Advanced Air Mobility: Foundations, Technical Advances, and an Approach to Travel Demand Modeling," IS 792 – Advanced Topics in Information Systems, UMBC, Nov 20, 2025.

PROFESSIONAL SERVICE & LEADERSHIP

Professional Leadership and Conference Service

- Vice-Chair, IEEE Baltimore Section, 2026.
- Co-Chair, IEEE Baltimore Technical Colloquium & Professional Development Conference, 2026.
- Conference Secretary, IEEE Baltimore Technical Colloquium & Professional Development Conference, 2025.
- IEEE Day Ambassador, IEEE Region 2, 2025–2026.
- Region 2 Lead, IEEEXtreme 18.0, 19.0, and 20.0.
- Technical Program Committee Member, 21st International Multi-Conference on Computing in the Global Information Technology (ICCGI 2026).

Editorial Roles

- Newsletter Editor, ACM Emerging Interest Group on Trustworthy AI and Autonomous Systems (EIGTRUST), 2026–Present.

Journal Peer Review Service

- IEEE Transactions on Intelligent Transportation Systems, 2022–Present.
- IEEE Transactions on Artificial Intelligence, 2023–Present.
- IEEE Transactions on Neural Networks and Learning Systems, 2024–Present.
- IEEE Transactions on Consumer Electronics, 2024–Present.
- IEEE SMC Magazine, 2024–Present.
- IEEE Internet of Things Journal, 2025–Present.
- Journal of Big Data, 2024–Present.
- Arabian Journal for Science and Engineering, 2025–Present.
- KSII Transactions on Internet and Information Systems, 2024–Present.

TECHNICAL EXPERTISE

Programming	Python, C, C++, PHP, SQL
AI and Machine Learning	PyTorch, TensorFlow, Scikit-learn, Deep Learning, Reinforcement Learning, Neurosymbolic AI, Symbolic Knowledge Distillation, Interpretable Machine Learning
Mathematical Optimization	Mixed-Integer Optimization, Distributed Optimization, ADMM, Genetic Algorithms, Learning-Based Optimization
Computational Tools	Gurobi, Pyomo, MATLAB, Git, MySQL, CircuitLab
Modeling and Simulation	Transportation Demand Forecasting, Gravity Models, Scenario-Based Simulation, Fleet Operations Modeling
Languages	English, Nepali, Hindi

PROFESSIONAL MEMBERSHIPS

- Senior Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, Association for Computing Machinery (ACM)
- Member, Phi Kappa Phi Honor Society
- Registered Engineer, Nepal Engineering Council